

1 Freeze the read plan

Before any outcome is read, draw the whole plan $\mathcal{A}$: arrival order, a 50/50 arm per arrival, re-selection on exhaustion. Every count $n_c(k)$ is a function of $\mathcal{A}$.



2 Bound each cell with its finite-population variance

Given $\mathcal{A}$, cell c is a without-replacement sample of size $n_c(k)$. Bennett with variance $nm(1-m)(N-n)/(N-1)$ bounds its log-MGF; an exhausted cell contributes 0.



3 One Chernoff width per policy difference

Sum the cell bounds along the direction (π' vs. $\hat{\pi}$) and optimise λ : errors combine like an ℓ_2 norm of standard errors, whereas rectangles add per-cell radii (ℓ_1).



4 Certify only at $K = 20$ fixed checkpoints

Certify budget q once $\hat{\Delta} + w \leq \varepsilon$ for every feasible challenger. A union over (problem, policy, checkpoint) and exact-hypergeometric variance boxes give FWER $\leq \delta$.